Ricardo Rei

London, United Kigdom ricardo.rei@outlook.com +44 7771 195960 ricardorei.github.io
ricardo-rei Scholar - Ricardo Rei

I am Head of AI Research at Sword Health, leading work on multimodal foundational models for the healthcare sector. I hold a Ph.D. from Instituto Superior Técnico, in collaboration with Carnegie Mellon University. Previously, I was a Senior Research Scientist at Unbabel, where I led research on multilingual LLMs, with a focus on model evaluation, post-training techniques, and deployment.

Education

Doctoral Studies (PhD) in Artificial Intelligence *Lisbon*
Instituto Superior Técnico & Carnegie Mellon University *Sept 2020 – April 2024*
During my Ph.D., I focused on machine translation (MT) evaluation and developed COMET, which has become the de facto standard metric for MT evaluation and has inspired a wide range of follow-up work on model-based evaluation. In addition, I contributed to research in related areas such as NLP efficiency, interpretability, and decoding.

- **Thesis title**:Robust, Interpretable and Efficient MT Evaluation with Fine-tuned Metrics
- **Advisors:** Luisa Coheur (IST) and Alon Lavie (CMU)
- Graduated with the distinction of *Summa cum laude*

Master of Science in Information Systems and Computer Engineering *Lisbon*
Instituto Superior Técnico *2017 – 2019*

- Conducted M.Sc. thesis on dialog systems, comparing generative and retrieval-based approaches.
- Specialized in Artificial Intelligence.
- Graduated with distinction (18/20).

Graduation in Computer Science and Engineering *Lisbon*
Instituto Superior Técnico *2013 – 2017*

Graduated with Merit, with top marks in core algorithm and programming courses.

Experience

Head of AI Research *London*
Sword Health *Jul 2025 – Present*

- Leading work on multimodal foundational models for the healthcare sector.

Senior Research Scientist *Lisbon and London*
Unbabel *Oct 2018 – Jun 2025*

- Started at Unbabel as an intern and progressed to Senior Research Scientist, becoming a key member of the research team both as an individual contributor and as the main point of contact between research, product, and sales teams, aligning work across groups to ensure successful deployment of Tower LLM and quality estimation models to clients.
- Principal researcher and developer of Tower LLM, a suite of multilingual large language models optimized for translation tasks. Tower+ achieve a Pareto frontier between general multilingual capabilities and translation performance. Earlier Tower models (Tower v2) won the WMT shared task on large language model alignment.
- Principal researcher and developer of the COMET framework and its derivative models. Since its initial release, COMET models have become go-to metrics for MT evaluation and quality estimation, with significant impact in both academia and industry.
- Core developer on the EuroLLM project. I worked on pretraining by designing and applying quality filtering

techniques, contributed to the annealing phase, and led all aspects of post-training for two EuroLLM models (1.7B, 9B and 22B parameters). These models were trained from scratch on 4 trillion tokens across 35 languages and are competitive with Mistral and Meta models of similar size for European languages.

- Senior advisor on the development of EuroBERT, a family of multilingual encoders with up to 2B parameters, trained on 4 trillion tokens. EuroBERT incorporates several recent innovations from decoder-only models, bringing those advancements into the encoder-based paradigm.

Teaching Assistant *Lisbon, Portugal*
Instituto Superior Técnico *Sep 2021 – Feb 2023*

- Teaching Assistant for the MSc course in Natural Language Processing and the BSc course in Logic Programming at Instituto Superior Técnico since 2021.
- Received the "Outstanding Teacher" award from the Computer Science Department (DEI) at IST in both 2021/2022 and 2022/2023 academic years.

Visiting Student *Pittsburgh, United States*
Carnegie Mellon University *May 2022 – Sep 2022*
As part of my PhD program, I was a visiting student at CMU's Language Technologies Institute (LTI), where I worked alongside my co-advisor, Alon Lavie.

Invited Lecturer *Lisbon, Portugal*
NOVA IMS Information Management School *Jan 2019 – Dec 2021*
Invited teacher at NOVA IMS for two prominent programs: Data Science and Advanced Analytics MSc: Professor of **Text Mining**, Information Management MSc: Professor of **Text Analytics**

Students and Mentorship

Currently, I am **co-advising 2 PhD students**:

- **José Pombal**: Working on LLM evaluation and post-training. He is one of the main authors of Tower LLM.
- **Helena Wu**: Developing Culturally-Informed AI Agent and Personae.

Co-advised M.Sc. thesis from Instituto Superior Técnico (Finished):

- NLP-Telescope: Extending MT-Telescope for Natural Language Processing Models (2023) – Rita Oliveira
- Energy-Efficiency in Natural Language Processing (2023) – Pedro Ramos
- Robust Machine Translation Evaluation Focusing on Critical Errors (2022) – Duarte Alves
- Sentiment-Aware Conversational Agent (2021) – Isabel Dias

Co-advised M.Sc. thesis from NOVA IMS (Finished):

- Multi-language Email Classification Using Transfer learning (2022) - Mário Sousa
- Document Clustering as an approach to template extraction (2022) - André Rodrigues
- Multilingual Email Zoning: Segmenting Multilingual Email Text Into Zones (2021) - Bruno Jardim
- Classification problem in real estate corpora: Furniture detection in real estate listings (2021) - Alexandra Ordina

Selected Publications

For a complete list, please visit my Google Scholar profile. * = (co-)first author

Tower+: Bridging Generality and Translation Specialization in Multilingual LLMs (2025). **R. Rei***, N.M. Guerreiro*, J. Pombal*, J. Alves*, P. Teixeirinha, A. Farajian, A.F.T Martins. **ArXiv**.

Zero-shot Benchmarking: A Framework for Flexible and Scalable Automatic Evaluation of Language Models (2025). J. Pombal, N.M. Guerreiro, **R. Rei**, A.F.T Martins. **ArXiv**.

EuroBERT: Scaling Multilingual Encoders for European Languages (2025). N. Boizard*, H. Gisserot-Boukhlef*, D.M. Alves*, A. Martins, A. Hammal, C. Corro, C. Hudelot, E. Malherbe, E. Malaboeuf, F. Jourdan, G. Hautreux, J. Alves, K. El-Haddad, M. Faysse, M. Peyrard, N.M. Guerreiro, P. Fernandes, **R. Rei**, P. Colombo. **ArXiv**.

EuroLLM: Multilingual language models for europe (2025). P.H. Martins*, P. Fernandes*, J. Alves*, N.M.

Guerreiro*, **R. Rei***, D.M. Alves, J. Pombal, A. Farajian, M. Faysse, M. Klimaszewski, P. Colombo, B. Haddow, J.G.C. de Souza, A. Birch, A.F.T. Martins. **ArXiv**.

Tower v2: Unbabel-IST 2024 Submission for the General MT Shared Task (2024). **R. Rei***, J. Pombal*, N.M. Guerreiro*, J. Alves, P.H. Martins, P. Fernandes, H. Wu, T. Vaz, D. Alves, A. Farajian, S. Agrawal, A. Farinhas, J.G.C. De Souza, A.F.T. Martins. **WMT 24**. 🏅 Winners of the WMT24 General Translation Shared Task

Tower: An Open Multilingual Large Language Model for Translation-Related Tasks (2024). D. Alves*, J. Pombal*, N.M. Guerreiro*, P.H. Martins, J. Alves, A. Farajian, B. Peters, **R. Rei**, P. Fernandes, S. Agrawal, P. Colombo, J.G.C. Sousa, A.F.T. Martins. **COLM**. Spotlight Paper.

QUEST: Quality-Aware Metropolis-Hastings Sampling for Machine Translation (2024). G.R.A Faria, S. Agrawal, A. Farinhas, **R. Rei**, J.G.C de Souza, A.F.T Martins **NeurIPS 2024**.

xCOMET: Transparent Machine Translation Evaluation through Fine-grained Error Detection (2024). N.M. Guerreiro*, **R. Rei***, D. Stigt, L. Coheur, P. Colombo, A.F.T Martins **TACL**. 🏅 Winners of the WMT23 and WMT24 Metrics Shared Task

Scaling up CometKiwi: Unbabel-IST 2023 Submission for the Quality Estimation Shared Task (2023). **R. Rei***, N.M. Guerreiro*, J. Pombal, D. van Stigt, M. Treviso, L. Coheur. J.G.C Souza, A.F.T Martins. **WMT 23**. 🏅 Winners of the WMT23 QE Shared Task

CometKiwi: IST-Unbabel 2022 Submission for the Quality Estimation Shared Task (2022). **R. Rei***, M. Treviso*, N.M. Guerreiro*, C. Zerva*, A. Farinha, C. Maroti, J. Souza, T. Glushkova, D. Alves, A. Lavie, L. Coheur, A.F.T. Martins **WMT 22**. 🏅 Winners of the WMT22 QE Shared Task

COMET: A Neural Framework for MT Evaluation (2022). **R. Rei**, C. Stewart, A.C. Farinha, A Lavie **EMNLP 2020**.

Selected Talks

Keynote at WMT 24: "What Makes MT Research Special in the LLM Age?" URL: https://www2.statmt.org/wmt24/program.html

Keynote at Sword AI Summit 24: "Tower LLM: An Open Multilingual LLM for Translation" URL: https://aisummit.swordhealth.com/

Keynote at First International Workshop on Knowledge-Enhanced Machine Translation "TowerLLM: Improving Translation Quality through Prompting with Terminology and Translation Guidelines" URL: https://kemt2024.wixsite.com/home

Keynote at MT Marathon 2024 "Large Language Models for Translation" URL: https://ufal.mff.cuni.cz/mtm24/programme.html

Keynote at MT Marathon 2022 "Machine Translation Quality Estimation" URL: https://ufal.mff.cuni.cz/mtm22/programme.html

Awards

🎖 **Best 2024 European Ph.D. Thesis in Machine Translation:** 2024 Anthony C Clarke Award

🎖 **PhD student of the year 2023** INESC-ID Award

🏅 **Best paper award: European Association for Machine Translation** (EAMT 2022)

Skills

Languages: Portuguese (native), English (fluent, fully proficient)